INCH-POUND

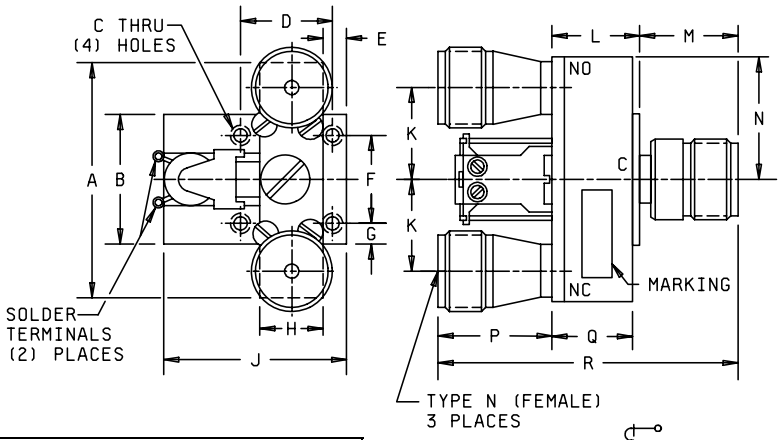
MIL-DTL-3928/10E
w/AMENDMENT 1
8 January 2008
SUPERSEDING
MIL-DTL-3928/10E
15 August 2001

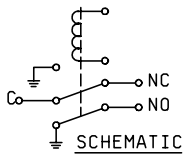
DETAIL SPECIFICATION SHEET

# SWITCHES, RADIO-FREQUENCY TRANSMISSION LINE (COAXIAL) (ELECTRICALLY OPERATED) CLASS 1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Requirements for acquiring the switch described herein shall consist of this specification and MIL-DTL-3928.




| |
|---|
| Position 1. (De-energized or failsafe position), C connects to NC connector. |
| Position 2. (Energized position), C connects to NO connector. |

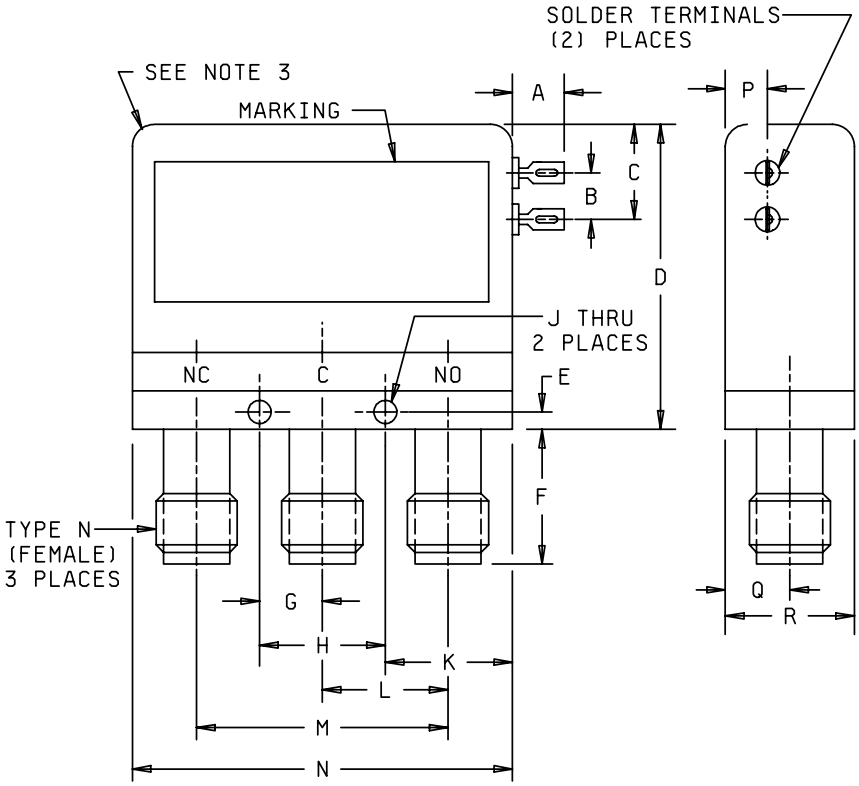
| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Letter | Inches | | Millimeters | | Letter | Inches | | Millimeters | |
| | Max | Min | Max | Min | | Max | Min | Max | Min |
| A | 2.03 | 1.97 | 51.6 | 50.0 | J | 1.510 | 1.490 | 38.35 | 37.85 |
| B | 1.135 | 1.115 | 28.83 | 28.32 | K | .792 | .772 | 20.12 | 19.61 |
| C | .112-40 UNC-2B dia | | 2.84-40 UNC-2B dia | | L | .604 | .584 | 15.34 | 14.83 |
| D | .697 | .677 | 17.70 | 17.20 | M | .681 | .661 | 17.30 | 16.79 |
| E | .104 | .084 | 2.64 | 2.13 | N | 1.03 | .97 | 26.2 | 24.6 |
| F | .885 | .865 | 22.48 | 21.97 | P | .775 | .755 | 19.68 | 19.18 |
| G | .135 | .115 | 3.43 | 2.92 | Q | .510 | .490 | 12.95 | 12.45 |
| H | .510 | .490 | 12.95 | 12.45 | | | | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (in millimeters) are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 1. PINs M3928/10-01, M3928/10-02, and M3928/10-03.



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Position 1. (De-energized or failsafe position),
C connects to NC connector.
Position 2. (Energized position),
C connects to NO connector.

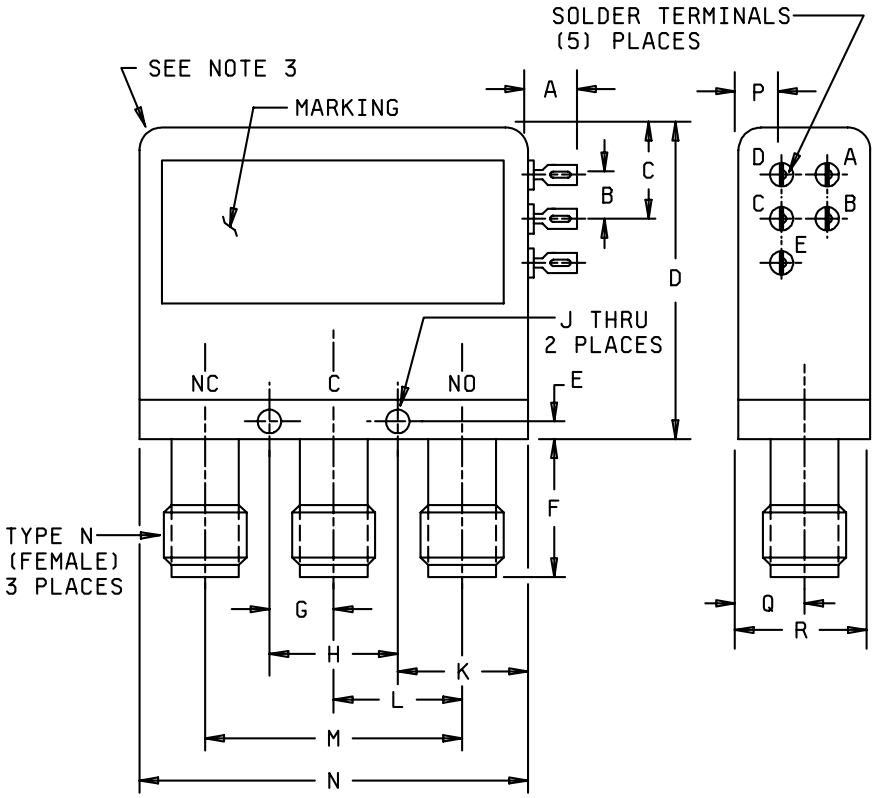
| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Inches | | Millimeters | | | Inches | | Millimeters | |
| Letter | Max | Min | Max | Min | Letter | Max | Min | Max | Min |
| A | .32 | --- | 8.1 | --- | J | .213 dia | .193 dia | 5.41 dia | 4.90 dia |
| B | .38 | .32 | 9.7 | 8.1 | K | .880 | .860 | 22.35 | 21.84 |
| C | .73 | .67 | 18.5 | 17.0 | L | .895 | .875 | 22.73 | 22.23 |
| D | 2.2 | --- | 56 | --- | M | 1.780 | 1.760 | 45.21 | 44.70 |
| E | .210 | .190 | 5.33 | 4.83 | N | 2.71 | --- | 68.8 | --- |
| F | .63 | .57 | 16.0 | 14.5 | P | .27 | .21 | 6.9 | 5.3 |
| G | .452 | .432 | 11.48 | 10.97 | Q | .56 | .50 | 14.2 | 12.7 |
| H | .895 | .875 | 22.73 | 22.23 | R | 1.09 | --- | 27.7 | --- |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (in millimeters) are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 2. PINs M3928/10-04 and M3928/10-09.



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Position 1. (De-energized or failsafe position), C connects to NC connector.
Position 2. (Energized position), C connects to NO connector.

| Actuator and ind. ckt terminals | |
|---|---|
| Solenoid common | A |
| Solenoid (position 2) +28 V dc | B |
| Ind. ckt common | C |
| Ind. ckt (position 1) | D |
| Ind. ckt (position 2) | E |

| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Inches | | Millimeters | | | Inches | | Millimeters | |
| Letter | Max | Min | Max | Min | Letter | Max | Min | Max | Min |
| A | .32 | --- | 8.1 | --- | J | .213 dia | .193 dia | 5.41 dia | 4.90 dia |
| B | .38 | .32 | 9.7 | 8.1 | K | .880 | .860 | 22.35 | 21.84 |
| C | .73 | .67 | 18.5 | 17.0 | L | .895 | .875 | 22.73 | 22.23 |
| D | 2.2 | --- | 56 | --- | M | 1.780 | 1.760 | 45.21 | 44.70 |
| E | .210 | .190 | 5.33 | 4.83 | N | 2.71 | --- | 68.8 | --- |
| F | .63 | .57 | 16.0 | 14.5 | P | .27 | .21 | 6.9 | 5.3 |
| G | .452 | .432 | 11.48 | 10.97 | Q | .555 | .535 | 14.10 | 13.59 |
| H | .895 | .875 | 22.73 | 22.23 | R | 1.09 | --- | 27.7 | --- |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (in millimeters) are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 3. PIN M3928/10-05.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.

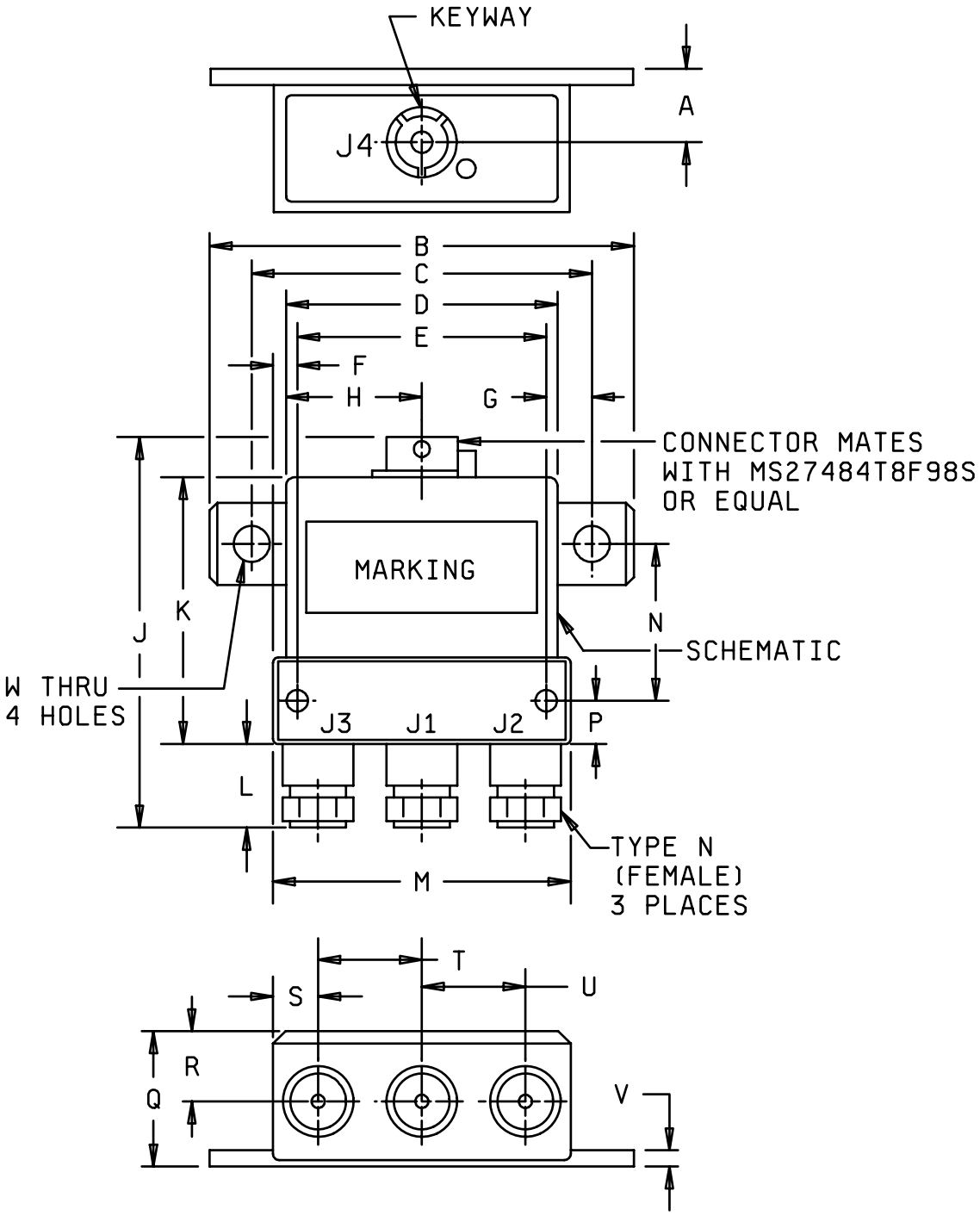
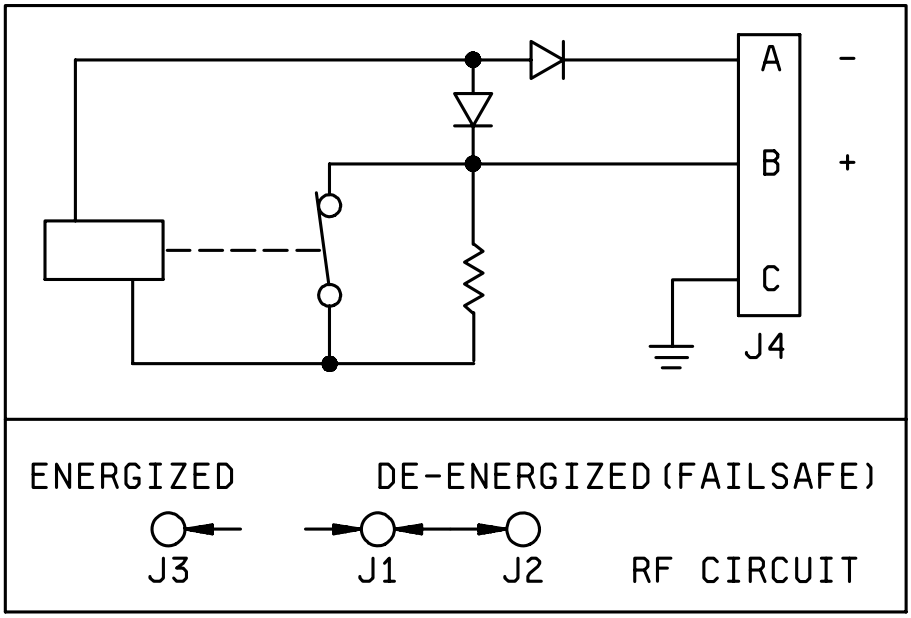



FIGURE 4. PIN M3928/10-06.



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SCHEMATIC




| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Inches | | Millimeters | | | Inches | | Millimeters | |
| Letter | Max | Min | Max | Min | Letter | Max | Min | Max | Min |
| A | .59 | .53 | 15.0 | 13.5 | M | 2.72 | 2.66 | 69.1 | 67.6 |
| B | 3.47 | 3.41 | 88.1 | 86.6 | N | 1.545 | 1.535 | 39.2 | 39.0 |
| C | 2.945 | 2.935 | 74.80 | 74.55 | P | .39 | .33 | 9.9 | 8.4 |
| D | 2.53 | 2.47 | 64.3 | 62.7 | Q | 1.15 | 1.09 | 29.2 | 27.7 |
| E | 2.395 | 2.385 | 60.83 | 60.58 | R | .59 | .53 | 15.0 | 13.5 |
| F | .18 | .12 | 4.6 | 3.0 | S | .50 | .44 | 12.7 | 11.2 |
| G | .30 | .24 | 7.6 | 6.1 | T | .873 | .867 | 22.17 | 22.02 |
| H | 1.28 | 1.22 | 32.5 | 31.0 | U | .873 | .867 | 22.17 | 22.02 |
| J | 3.66 | --- | 93.0 | --- | V | .10 | .04 | 2.5 | 1.0 |
| K | 2.63 | --- | 66.8 | --- | W | .185 | .175 | 4.70 | 4.44 |
| L | .69 | .63 | 17.5 | 16.0 | | | | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (in millimeters) are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 4. PIN M3928/10-06 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.

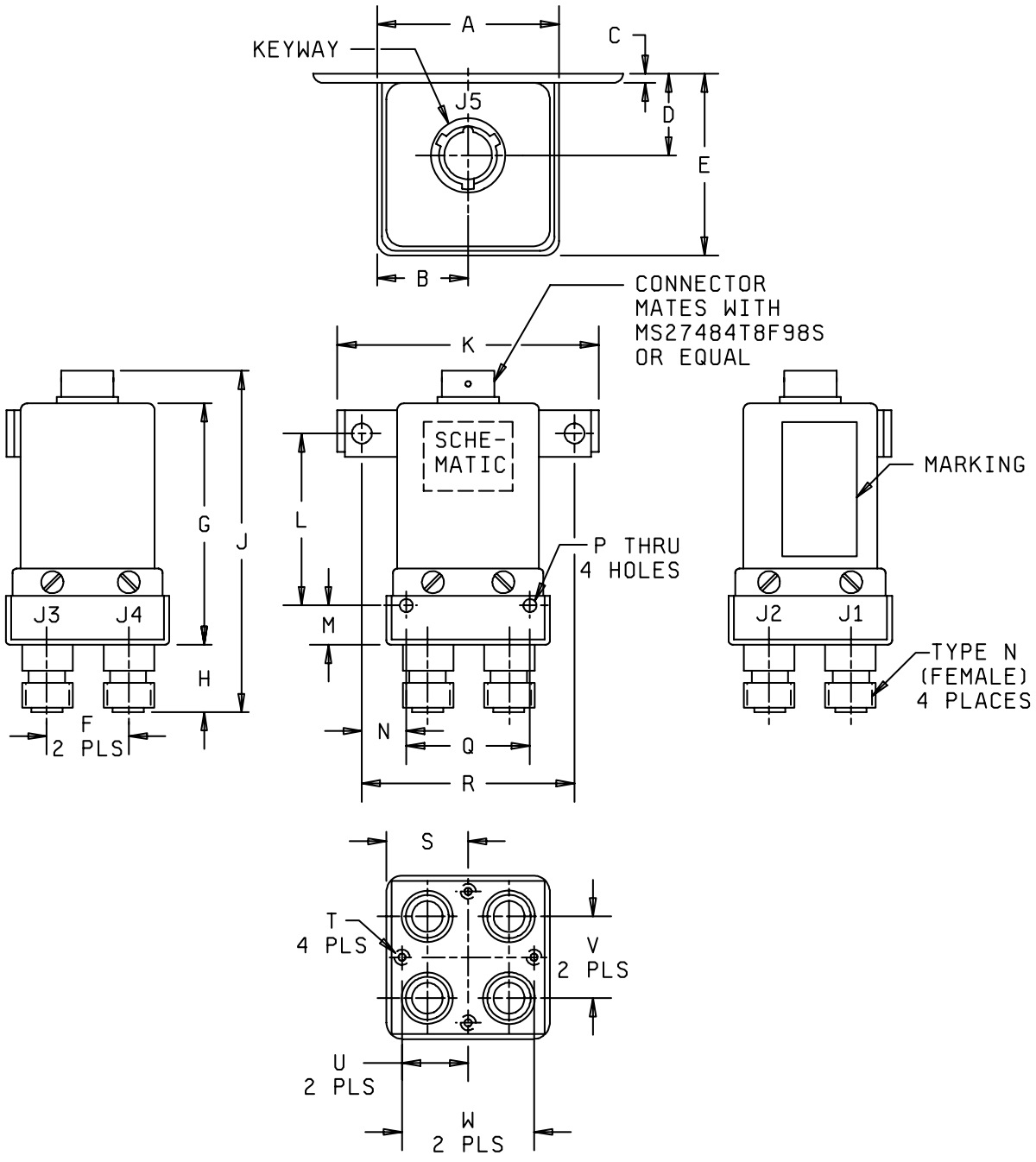
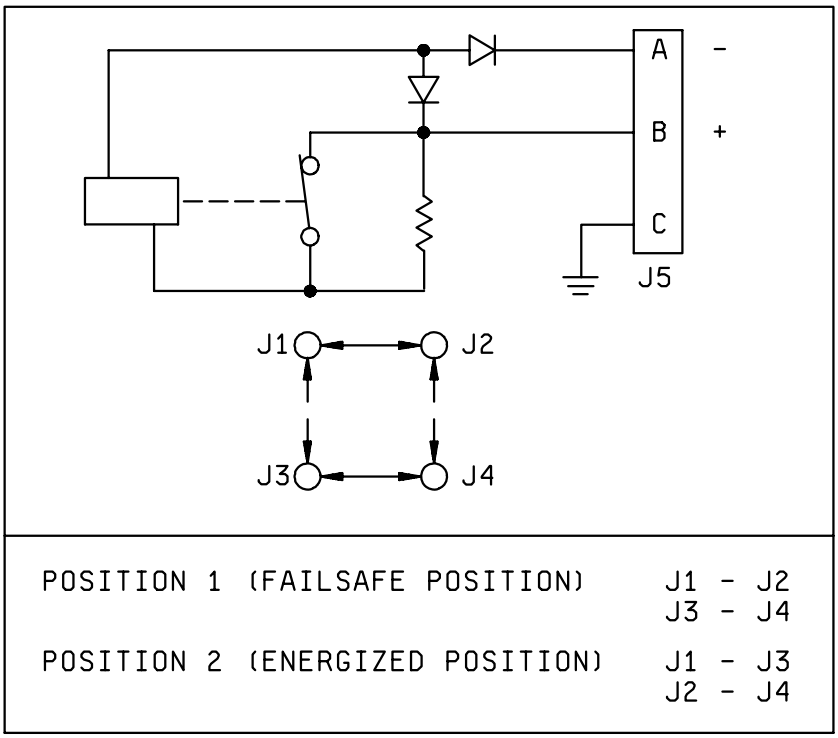



FIGURE 5. PIN M3928/10-07.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.

SCHEMATIC




POSITION 1 (FAILSAFE POSITION) J1 - J2
J3 - J4

POSITION 2 (ENERGIZED POSITION) J1 - J3
J2 - J4

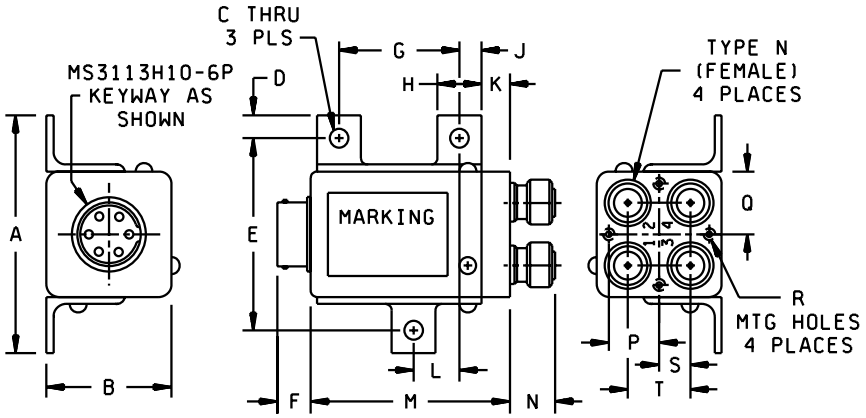
| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Inches | | Millimeters | | | Inches | | Millimeters | |
| Letter | Max | Min | Max | Min | Letter | Max | Min | Max | Min |
| A | 2.00 | --- | 50.8 | --- | M | .40 | .33 | 10.2 | 8.4 |
| B | 1.10 | .90 | 27.9 | 22.9 | N | .37 | .31 | 9.4 | 7.9 |
| C | .107 | .097 | 2.72 | 2.46 | P | .185 dia | .175 dia | 4.70 dia | 4.45 dia |
| D | 1.10 | .90 | 27.9 | 22.9 | Q | 1.567 | 1.557 | 39.80 | 39.55 |
| E | 2.00 | --- | 50.8 | --- | R | 2.255 | 2.245 | 57.28 | 57.02 |
| F | 1.03 | .97 | 26.2 | 24.6 | S | 1.010 ref | .990 ref | 25.65 ref | 25.15 ref |
| G | 3.28 | 3.22 | 83.3 | 81.8 | T | .164-32 UNC-2B | | 4.17-32 UNC-2B | |
| H | .69 | .63 | 17.5 | 16.0 | | x .37 DP | (No. 8) | x 9.40 DP | (No. 8) |
| J | 4.40 | --- | 111.8 | --- | U | .817 | .807 | 20.75 | 20.50 |
| K | 2.78 | 2.72 | 70.6 | 69.1 | V | 1.03 | .97 | 26.2 | 24.6 |
| L | 2.40 | 2.34 | 61.0 | 59.4 | W | 1.630 | 1.620 | 41.40 | 41.15 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (in millimeters) are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

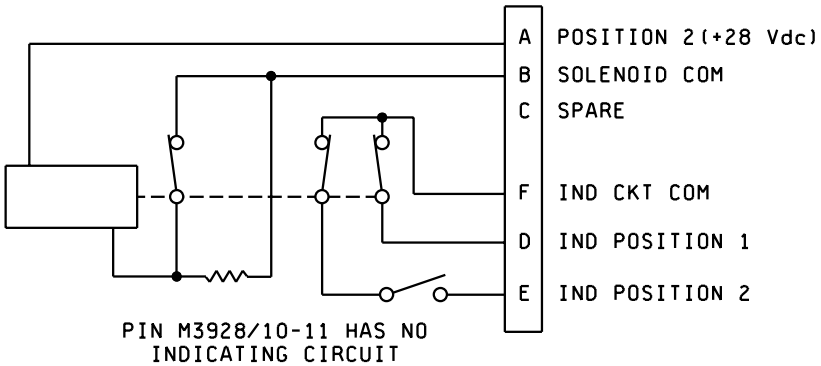
FIGURE 5. PIN M3928/10-07 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:16Z
Check the source to verify that this is the current version before use.

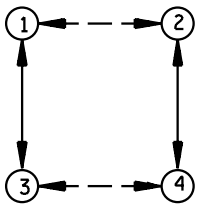



FAILSAFE SCHEMATIC FOR PIN M3928/10-08 AND 11




PIN M3928/10-11 HAS NO
INDICATING CIRCUIT

RF CIRCUIT




POSITION 1 (FAILSAFE POSITION) 1-3,2-4

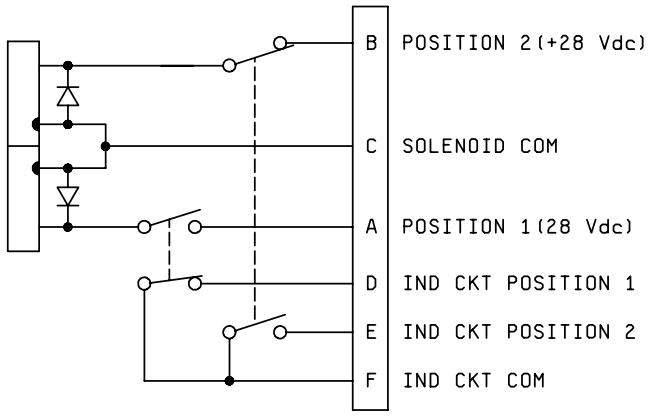
POSITION 2 (ENERGIZED POSITION) 1 CONNECTS
TO 2 AND 3 CONNECTS TO 4

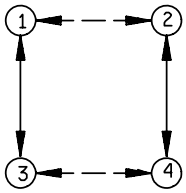
FIGURE 6. PINs M3928/10-08, M3928/10-10, and M3928/10-11.



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LATCHING SCHEMATIC FOR PIN M3928/10-10




POSITION 1 ⟷ 1-3,2-4
POSITION 2 ⟷ 1-2,3-4

RF CIRCUIT

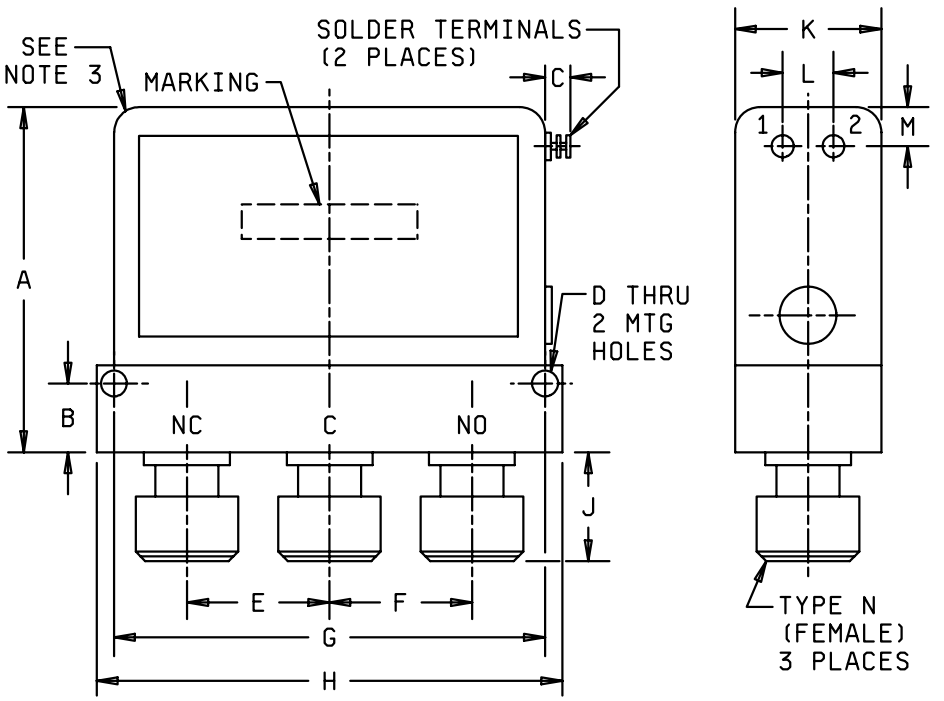
| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Letter | Inches | | Millimeters | | Letter | Inches | | Millimeters | |
| | Max | Min | Max | Min | | Max | Min | Max | Min |
| A | 3.28 | 3.22 | 83.3 | 81.8 | L | .753 | .747 | 19.13 | 18.97 |
| B | 1.78 | --- | 45.2 | --- | M | 2.60 | --- | 66.0 | --- |
| C | .198 dia | .192 dia | 5.03 dia | 4.88 dia | N | .65 | --- | 16.5 | --- |
| D | .35 | .29 | 8.9 | 7.4 | P | .730 | .710 | 18.54 | 18.03 |
| E | 2.610 | 2.590 | 66.29 | 65.79 | Q | .885 | .865 | 22.48 | 21.97 |
| F | .49 | --- | 12.4 | --- | R | .138-32 UNC-2B x .25 DP (No. 6) | | 3.50-32 UNC-2B x 6.4 DP (No. 6) | |
| G | 1.510 | 1.490 | 38.35 | 37.85 | | | | | |
| H | .63 | .57 | 16.0 | 14.5 | S | .452 | .432 | 11.48 | 10.97 |
| J | .33 | .27 | 8.4 | 6.9 | T | .895 | .875 | 22.73 | 22.23 |
| K | .28 | .22 | 7.1 | 5.6 | | | | | |

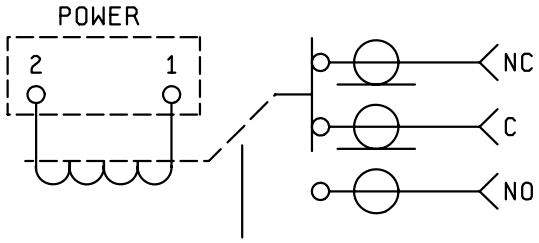
NOTES:
1. Dimensions are in inches.
2. Metric equivalents (in millimeters) are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 6. PINs M3928/10-08, M3928/10-10, and M3928/10-11 - Continued.



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Position 1. (De-energized or failsafe position), C connects to NC connector.
Position 2. (Energized position), C connects to NO connector.

| Dimensions | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Letter | Inches | | Millimeters | | Letter | Inches | | Millimeters | |
| | Max | Min | Max | Min | | Max | Min | Max | Min |
| A | 1.80 | --- | 45.7 | --- | G | 2.53 | 2.47 | 64.3 | 62.7 |
| B | .390 | .370 | 9.91 | 9.40 | H | 2.78 | 2.72 | 70.6 | 69.1 |
| C | .32 | --- | 8.1 | --- | J | .63 | .57 | 16.0 | 14.5 |
| D | .166 | .156 | 4.22 | 3.96 | K | .93 | .87 | 23.6 | 22.1 |
| E | .890 | .870 | 22.61 | 22.10 | L | .28 | .22 | 7.1 | 5.6 |
| F | .890 | .870 | 22.61 | 22.10 | M | .23 | .17 | 5.8 | 4.3 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (in millimeters) are given for general information only (1.00 inch = 25.4 mm).
3. Corners of case may be round or square.
4. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.

FIGURE 7. PIN M3928/10-12.



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TABLE I. Electrical and performance characteristics.

| PIN M3928/10- | Fig. No. | Housing | Config. | Manual or remote | Solenoid or motor | Fail-safe or latching | Frequency range dc to GHz | VSWR | Insertion loss (dB) | Isolation (dB) | Switching time (ms) | Position indicating circuit | Life cycles x 1000 | Operating voltage | Pick-up voltage | Drop-out voltage | Operating current (A) 1/ | Holding current fail-safe type 1/ (A) | Power and indicator connector | Weight (oz) 2/ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | max | max | min | max | | | nom | max | min | max | | | max |
| 01N, S | 1 | O | 1P2T | R | S | F | 5.2 | 1.4:1 | 0.48 | 25 | 20 | None | 1000 | 110 V dc | 84 V dc | 10 V dc | 0.2 | 0.2 | Solder terminals | 8 (227) |
| 02N, S 3/ | 1 | O | 1P2T | R | S | F | 5.2 | 1.4:1 | 0.48 | 25 | 20 | None | 1000 | 28 V dc | 18 V dc | 3 V dc | 0.2 | 0.2 | Solder terminals | 8 (227) |
| 03N, S 3/ | 1 | O | 1P2T | R | S | F | 5.2 | 1.4:1 | 0.48 | 25 | 20 | None | 1000 | 115 V ac | 93 V ac | 5 V ac | 0.2 | 0.2 | Solder terminals | 8 (227) |
| 04N, S | 2 | E | 1P2T | R | S | F | 12.4 | 1.5:1 | 0.5 | 60 | 50 | None | 100 | 28 V dc | 18 V dc | 3 V dc | 0.8 | 0.4 | Solder terminals | 8 (227) |
| 05N, S | 3 | E | 1P2T | R | S | F | 12.4 | 1.5:1 | 0.5 | 60 | 50 | Yes | 100 | 28 V dc | 18 V dc | 3 V dc | 0.8 | 0.4 | Solder terminals | 8 (227) |
| 06N, S | 4 | H | 1P2T | R | S | F | 12.4 | 1.5:1 5/ | 0.5 6/ | 25 7/ | 20 30 4/ | None | 50 | 28 V dc | 20 V dc | 14 V dc | 1.0 | 0.5 | see Fig 4 | 12 (342) |
| 07N, S | 5 | H | TR | R | S | F | 12.4 | 1.5:1 5/ | 0.5 6/ | 25 7/ | 20 30 4/ | None | 50 | 28 V dc | 20 V dc | 14 V dc | 1.5 | 0.6 | see Fig 5 | 16 (456) |
| 08N, S | 6 | H | TR | R | S | F | 12.4 | 1.5:1 5/ | 0.5 6/ | 60 7/ | 30 | Yes | 100 | 28 V dc | 20 V dc | 14 V dc | 1.0 | 0.3 | see Fig 6 | 16 (456) |
| 09N, S | 2 | E | 1P2T | R | S | F | 12.4 | 1.5:1 | 0.5 | 60 | 50 | None | 100 | 115 V dc | 95 V ac | 5 V ac | 0.12 | .12 | Solder terminals | 8 (227) |
| 10N, S | 6 | H | TR | R | S | L | 12.4 | 1.5:1 5/ | 0.5 6/ | 60 | 20 | Yes | 100 | 28 V dc | 20 V dc | --- | 0.64 | --- | see Fig 6 | 12 (342) |
| 11N,S | 6 | H | TR | R | S | F | 12.4 | 1.5:1 | 0.5 | 60 | 20 | None | 100 | 28 V dc | 20 V dc | 14 V dc | 1.5 | .25 | see Fig 6 | 12 (342) |
| 12N, S | 7 | E | 1P2T | R | S | F | .96 to 1.215 | 1.2:1 | 0.3 | 60 | 30 | None | 100 | 28 V dc | 20 V dc | 14 V dc | 0.2 | 0.2 | Solder terminals | 8 (227) |

1/ At nominal operating voltage and 20°C.

2/ Mass in grams in parentheses.

3/ Open port is grounded.

4/ At low and high temperature extremes.

5/ The VSWR at .225-.400 GHz for dash numbers 06 and 07 is 1.1:1. The VSWR at .100-.500 GHz for dash number 08 is 1.15:1. The VSWR at 4 GHz for dash number 10 is 1.25:1.

6/ The insertion loss at .225-.400 GHz for dash numbers 06 and 07 is 0.1 dB. The insertion loss at .100-.500 GHz for dash number 08 is 0.15 dB. The insertion loss at 4 GHz for dash number 10 is 0.25 dB.

7/ The isolation at .225-.400 GHz for dash numbers 06 and 07 is 50 dB. The isolation at .100-.500 GHz for dash number 08 is 80 dB.



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REQUIREMENTS:

Dimensions and configurations: See figures 1 through 7.

RF connectors: Female connectors shall mate with type N connectors in accordance with MIL-PRF-39012/1.

Nominal impedance: 50 ohms.

Termination: Open.

RF power handling capability (average): 100 watts for dash numbers 01 through 11 and 280 watts for dash number 12.

RF connectors: Series N connectors shall mate with series N male connectors in accordance with MIL-PRF-39012/1.

Electrical and performance characteristics: See table I.

Vibration: Method I.

Operating temperature: -55 deg C to +85 deg C.

Part or Identifying Number (PIN): M3928/10- (dash number from table I).

TABLE II. PIN to type cross-reference.

| PIN M3928/10- | Type SA- /U |
|---|---|
| 01 | 1360 |
| 02 | 1361 |
| 03 | 1362 |
| 04 | --- |
| 05 | --- |
| 06 | --- |
| 07 | --- |
| 08 | --- |
| 09 | --- |
| 10 | --- |
| 11 | --- |
| 12 | --- |

Amendment notations. The margins of this specification sheet are marked with vertical line to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

NOTE

Referenced documents: In addition to MIL-DTL-3928, this specification sheet also references MIL-PRF-39012/1.



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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5985-2008-002)

Review activities:
Army - MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/.



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